

Problem Statement:

Cybersecurity Log Analyzer

Background

Every system that has a login page records each attempt as a log entry. When the same IP address fails to log in again and again, it is often a sign of a brute-force attack, where someone tries many passwords to break into an account. Checking such logs by hand is slow and easy to get wrong.

Problem

Write a Python program that reads login records, counts the successful and failed attempts for every IP address, and points out the IP addresses that look suspicious.

Input

1. The total number of logs (an integer).
2. For each log:

- **IP address:** a non-empty string, for example `192.168.1.10`
- **Status:** either `SUCCESS` or `FAILED` (not case sensitive)

Rules

- If the IP address is blank, show an error message and skip that record.
- If the status is not `SUCCESS` or `FAILED`, show an error message and skip that record.
- An IP address is **suspicious** if it has **3 or more failed** login attempts.
- If the total failed logins are **more than** the total successful logins, show a warning. Otherwise, show that the activity looks normal.

Output

The program should print three sections:

1. **Login Report:** successful and failed login counts for each IP address.
2. **Suspicious IPs:** every IP address with 3 or more failed attempts, or a message that none were found.

3. **Final Analysis:** total successful logins, total failed logins, and the warning or normal message.

Sample Input

```
Enter total number of logs: 5

Log 1
Enter IP address: 192.168.1.10
Enter status (SUCCESS/FAILED): FAILED

Log 2
Enter IP address: 192.168.1.10
Enter status (SUCCESS/FAILED): FAILED

Log 3
Enter IP address: 192.168.1.10
Enter status (SUCCESS/FAILED): FAILED

Log 4
Enter IP address: 10.0.0.5
Enter status (SUCCESS/FAILED): SUCCESS

Log 5
Enter IP address: 10.0.0.5
Enter status (SUCCESS/FAILED): SUCCESS
```

Sample Output

```
--- LOGIN REPORT ---

Successful Logins:
10.0.0.5 -> 2

Failed Logins:
192.168.1.10 -> 3

--- SUSPICIOUS IPS ---
192.168.1.10 -> 3 failed attempts

--- FINAL ANALYSIS ---
Total Successful Logins: 2
Total Failed Logins: 3
Warning: High number of failed login
attempts!
```

Approach

- Use two dictionaries, `success` and `failed`, where the key is the IP address and the value is the number of attempts.
- Split the work into four functions:

Function	Job
<code>get_log_data()</code>	Take input, validate it, and build the two dictionaries

Function	Job
<code>print_report()</code>	Print the per-IP login report
<code>check_suspicious_ips()</code>	Find IPs with 3 or more failed attempts
<code>calculate_totals()</code>	Add up all logins and print the final analysis

Concepts Used

Functions, dictionaries, loops, conditional statements, input validation, and data aggregation.